

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

Electron Configuration

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

A Retrieval

1. State the maximum number of electrons in the first, second and third shells for the first 20 elements as used in this course.
2. How many electrons can an s subshell and a p subshell hold?
3. What does the group number tell you about a Group 1 or Group 7 atom?

B Knowledge

4. Write shell configurations for N, Mg, Cl and K.
5. Write full subshell configurations for O, Al and Ca.
6. Explain why noble gases are chemically unreactive in terms of electron arrangement.

C Application

7. Write the configuration of O^{2-} , Al^{3+} and Cl^- .
8. A student writes calcium as 2,8,10. Correct the configuration and explain the error.
9. Explain, using electrons, why magnesium forms a $2+$ ion.

D Exam-style practice

10. Write the electron configuration of Mg^{2+} and explain why the ion is stable. [3]
11. Explain why oxygen typically forms a $2-$ ion. [3]
12. Identify the block (s, p or d) for sodium, fluorine and iron, and justify one of them. [3]

E Challenge

13. Write [Ne] shorthand configurations for phosphorus and for P^{3-} . Explain the relationship between them.